EXP NO: 6 EXPERIMENT TO UNDERSTAND FEATURE SCALING

AIM:

Understand the importance of feature scaling.

CODE:

```
import pandas as pd
from sklearn.preprocessing import StandardScaler, MinMaxScaler
data = {
 'Country': ['France', 'Spain', 'Germany', 'Spain', 'Germany',
 'France', 'Spain', 'France', None, 'France'],
 'Age': [44.0, 27.0, 30.0, 38.0, 40.0,
 35.0, None, 48.0, 50.0, 37.0],
 'Salary': [72000.0, 48000.0, 54000.0, 61000.0, None,
 58000.0, 52000.0, 79000.0, 83000.0, 67000.0],
 'Purchased': ['No', 'Yes', 'No', 'No', 'Yes',
 'Yes', 'No', 'Yes', 'No', 'Yes']
}
df = pd.DataFrame(data)
df['Age'] = df['Age'].fillna(df['Age'].mean())
df['Salary'] = df['Salary'].fillna(df['Salary'].mean())
X = df[['Age', 'Salary']]
standard_scaler = StandardScaler()
X_standardized = standard_scaler.fit_transform(X)
standardized_df = pd.DataFrame(
 X_standardized,
 columns=['Age', 'Salary']
)
print("\nStandardized Data:")
print(standardized_df)
minmax_scaler = MinMaxScaler()
X_normalized = minmax_scaler.fit_transform(X)
normalized_df = pd.DataFrame(
 X_normalized,
 columns=['Age', 'Salary']
)
print("\nNormalized Data:")
print(normalized_df)
```

OUTPUT:

```
Standardized Data:
        Age        Salary
0  0.758874  7.494733e-01
1 -1.711504 -1.438178e+00
2 -1.275555 -8.912655e-01
3 -0.113024 -2.532004e-01
4  0.177609  6.632192e-16
5 -0.548973 -5.266569e-01
6  0.000000 -1.073570e+00
7  1.340140  1.387538e+00
8  1.630773  1.752147e+00
9 -0.258340  2.937125e-01

Normalized Data:
        Age    Salary
0  0.739130  0.685714
1  0.000000  0.000000
2  0.130435  0.171429
3  0.478261  0.371429
4  0.565217  0.450794
5  0.347826  0.285714
6  0.512077  0.114286
7  0.913043  0.885714
8  1.000000  1.000000
9  0.434783  0.542857
```

RESULT:

Thus the code is executed successfully.